AGFM-Net: Angle-Guided Feature Modulation Network for HRRP Target Recognition

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***Abstract*—High-resolution range profile (HRRP) is widely used in radar target recognition due to its ability to characterize target scattering structures along the range dimension. However, HRRP signals are highly sensitive to aspect variations, leading to significant distribution shifts and performance degradation. This paper proposes an angle-guided deep learning framework for robust HRRP recognition. An angle-guided feature modulation (AGFM) mechanism incorporates observation angles into feature learning via channel-wise modulation parameters, enabling adaptive feature alignment under varying aspects. A convolutional–recurrent hybrid architecture captures both local scattering patterns and long-range dependencies, enhanced by a multilevel attention mechanism. Experiments on a multi-class HRRP dataset demonstrate that the proposed method outperforms CNN-based and hybrid baselines, achieving approximately 3% accuracy gain in six-class classification.**

***Index Terms*—HRRP, radar target recognition, angle-guided modulation, feature modulation, deep learning**

I. Introduction

Radar target recognition (RTR) plays a critical role in intelligent sensing systems for military surveillance, maritime monitoring, and autonomous navigation. Among various radar signal representations, high-resolution range profile (HRRP) has attracted considerable attention due to its capability to characterize target scattering structures. HRRP is formed by projecting scattering centers onto the radar line-of-sight (LOS), yielding a one-dimensional intensity distribution. As illustrated in Fig. 1, peaks in the profile correspond to dominant scatterers. Compared with 2D imaging techniques such as SAR and ISAR, HRRP offers a more compact yet informative representation.

However, HRRP signals are highly sensitive to aspect variations, causing significant distribution shifts and degraded recognition performance. To address this, we propose an angle-guided framework that integrates observation angles into feature learning via the AGFM mechanism. The main contributions of this paper are summarized as follows:

- To address the aspect sensitivity of HRRP signals, an angle-guided feature modulation (AGFM) module is proposed by explicitly incorporating angular information into feature extraction, enabling adaptive feature responses under varying observation conditions.

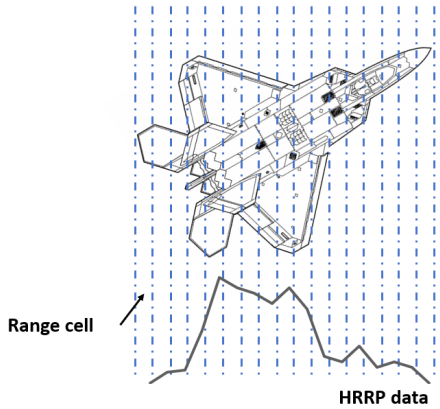

Fig. 1. Illustration of HRRP generation. The radar LOS projects scattering centers onto the range dimension.

- An attention mechanism is introduced to emphasize discriminative features and suppress irrelevant information, improving robustness and recognition accuracy.
- A hybrid framework combining AGFM and BiRNN is developed to jointly capture local scattering patterns and long-range dependencies.A simple yet effective angle-aware modulation strategy is designed to enhance feature consistency without increasing model complexity, achieving approximately 90% accuracy for corner reflectors and around 80% for ship targets.

II. Related Work

A. Traditional Methods

Early HRRP recognition relied on scattering center modeling, subspace analysis, and statistical frameworks. Matching pursuit (MP/OMP) extracted dominant scatterers combined with similarity measures for robust recognition [1]. Sparse methods including SRC and K-SVD improved reconstruction fidelity and discriminability [2], [3]. Subspace approaches (PCA, FA) projected HRRP data into low-dimensional spaces and paired with SVM classifiers [4]–[7]. Statistical models captured amplitude distributions via Gamma, AGC, and mix-

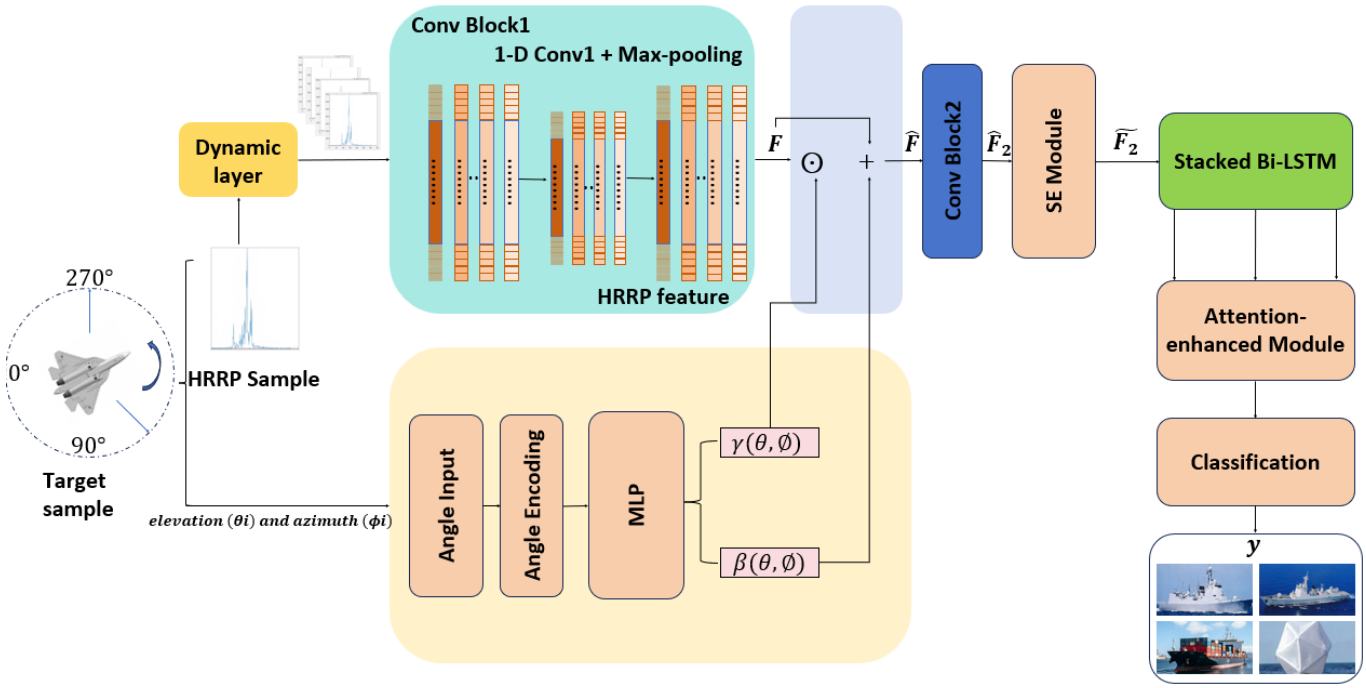

Fig. 2. Overall framework of AGFM-Net. Raw HRRP passes through a dynamic adjustment layer (DAL), then a hierarchical convolutional backbone. AGFM blocks after each convolutional stage incorporate angle-conditioned modulation. An SE block refines channel dependencies before BiLSTM layers and a multilevel attention module produce the final representation.

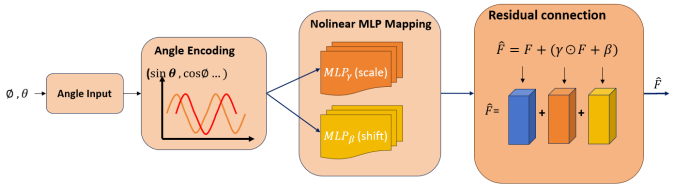

Fig. 3. Angle-guided feature modulation (AGFM) mechanism.

ture models [8]–[10], while HMMs modeled sequential range-cell dependencies [11]. However, predefined assumptions and linear modeling limited these methods under nonlinear conditions [9], [12].

B. Deep Learning Methods

CNNs with 1D convolutions, attention mechanisms, and multi-branch architectures significantly improved feature representation [13]–[18]. RNN-based methods (LSTM, GRU) modeled sequential HRRP dependencies and were extended via attention and CNN-RNN hybrids [19]–[23]. VAEs introduced generative modeling for latent distribution learning, with variants such as FDCVAE and CFCVAE improving discriminability [24]–[26]. Despite these advances, challenges including reliance on large labeled datasets, high computational cost, and limited generalization under distribution shift remain [27].

III. Methodology

Unlike conventional approaches that directly concatenate angle information with input features or implicitly encode it into convolution kernels, the proposed AGFM introduces angle awareness by modulating intermediate feature representations. This design enables the network to dynamically adjust its internal feature responses conditioned on the observation angle, thereby explicitly modeling the interaction between geometric configuration and electromagnetic scattering behavior. The overall AGFM framework consists of three tightly coupled components that progressively transform raw angle information into effective feature modulation signals.

A. Angle-Guided Feature Modulation

1) Angle Encoding: Directly using raw or normalized angle values may introduce artificial discontinuities and distort intrinsic geometric relationships between viewing directions. In particular, angles that are physically adjacent (e.g., 0° and 360°) may appear numerically distant, which can mislead the learning process. To address this, we adopt a trigonometric encoding scheme that maps angles onto a continuous manifold. Given azimuth θ and elevation ϕ:

$$e_\theta = [\sin\theta,\ \cos\theta], \quad e_\phi = [\sin\phi,\ \cos\phi] \tag{1}$$

$$e = [e_\theta,\ e_\phi] \in \mathbb{R}^4 \tag{2}$$

This encoding ensures continuity and preserves the geometric structure of angular information by projecting each angle onto the unit circle. Consequently, the Euclidean distance

between encoded vectors reflects true angular proximity, enabling smooth transitions across different observation aspects and facilitating stable downstream learning.

2) Angle Embedding via Nonlinear Mapping: While sinusoidal encoding captures the periodic structure of angles, its representational capacity remains limited for complex angle–feature interactions. The encoded vector e is therefore further transformed into a higher-dimensional latent representation through two independent lightweight MLPs:

$$\gamma = f_\gamma(e), \quad \beta = f_\beta(e), \quad \gamma, \beta \in \mathbb{R}^C \tag{3}$$

where C is the number of feature channels. This design allows the model to learn a nonlinear, channel-specific response to angle variations. Intuitively, different feature channels correspond to different scattering patterns or structural cues, and their sensitivities to viewing angles are not uniform. By learning separate scaling and shifting parameters, the network can selectively amplify or suppress specific feature channels according to the current observation geometry, yielding more discriminative and robust representations.

3) Angle-Guided Feature Modulation: Let $F \in \mathbb{R}^{C\times L}$ denote the intermediate feature map, where C is the number of channels and L the feature length. The AGFM module performs a channel-wise affine transformation:

$$\hat{F} = \gamma \odot F + \beta \tag{4}$$

where $\odot$ denotes channel-wise multiplication. Here, γ controls the relative importance of each channel, while β introduces an additive bias that shifts the feature response. This operation can be interpreted as a conditional feature recalibration mechanism, where the feature distribution is adaptively reshaped according to the observation angle, explicitly encoding angle-dependent scattering variations that are otherwise difficult to capture using standard convolutions.

To further improve training stability and preserve the original feature information, a residual formulation is adopted:

$$\hat{F} = F + (\gamma \odot F + \beta) \tag{5}$$

This residual design ensures that the modulation term acts as a corrective adjustment rather than a complete transformation, preventing potential degradation of useful baseline features in early training stages and facilitating gradient propagation. The AGFM module is inserted after intermediate convolutional layers:

$$F_l \rightarrow \text{Conv} \rightarrow \text{AGFM} \rightarrow F_{l+1} \tag{6}$$

where l denotes the layer index. Applying AGFM at early or intermediate layers is especially effective, as these layers capture local scattering structures and geometric patterns that are highly sensitive to target orientation, enabling better alignment of feature extraction with the underlying physical variations of HRRP signals.

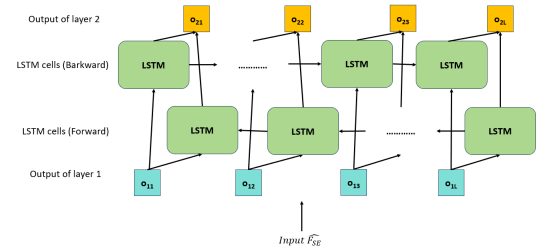

Fig. 4. Detailed architecture of the first bidirectional LSTM layer.

B. Sequential Feature Modeling via BiRNN

Following the second AGFM block, a squeeze-and-excitation (SE) module refines channel-wise features. Global average pooling (GAP) produces a compact channel descriptor, which passes through a bottleneck network (reduction ratio $r = 16$):

$$s = \sigma\Big(W_2 \cdot \text{ReLU}(W_1 \cdot \text{GAP}(\hat{F}_2))\Big), \quad \tilde{F}_2 = s \odot \hat{F}_2 \tag{7}$$

The SE block complements AGFM synergistically: while AGFM modulates features based on external angle information, SE recalibrates based on intrinsic feature content.

The recalibrated map $\tilde{F}_2$ is processed by a BiRNN to capture long-range dependencies in both directions:

$$\overrightarrow{\mathbf{h}}_t = f(\mathbf{x}_t, \overrightarrow{\mathbf{h}}_{t-1}), \quad \overleftarrow{\mathbf{h}}_t = f(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \tag{8}$$

$$\mathbf{h}_t = [\overrightarrow{\mathbf{h}}_t\,;\, \overleftarrow{\mathbf{h}}_t] \tag{9}$$

This bidirectional structure exploits contextual dependencies from both preceding and succeeding range cells, beneficial for capturing structural relationships among distributed scattering centers.

C. Attention-Enhanced Discrimination

An attention mechanism adaptively assigns importance weights to different range positions:

$$s_t = \mathbf{v}^T \tanh(\mathbf{W}_h \mathbf{h}_t + \mathbf{b}), \quad w_t = \frac{\exp(s_t)}{\sum_i \exp(s_i)} \tag{10}$$

$$\mathbf{h} = \sum_t w_t \mathbf{h}_t \tag{11}$$

This highlights salient scattering structures while suppressing noise and redundant information.

IV. Experiments

A. Dataset and Settings

The simulation dataset uses single-target scenarios under Sea Condition Level 1 in the X-band, generated via the shooting and bouncing rays (SBR) method. Elevation angles span 0°–90° and azimuth angles 0°–360°, with 160–180 frequency sampling points at 50× oversampling in HH polarization. The dataset includes destroyers, frigates, container ships, and three corner reflector configurations (Table II). All models are trained under identical settings using the Adam optimizer with learning rate decay. As shown in Fig. 5, elevation θ_i is

TABLE I
RECOGNITION ACCURACY (%) OF DIFFERENT MODELS FOR EACH CLASS

Model	Warship	Frigate	Containership	1×2 reflector	2×2 reflector	2×3 reflector	Overall
RNN	48.82	34.78	36.47	76.61	42.69	54.97	49.21
CNN	64.12	42.86	78.82	84.21	73.10	76.02	70.12
CNN-Bi-RNN	70.59	45.96	91.76	87.13	82.46	90.64	78.40
VAE	54.12	66.46	58.24	85.96	69.59	66.67	66.86
AGFM-Net (Ours)	68.82	65.84	90.59	90.06	84.21	88.89	**81.56**

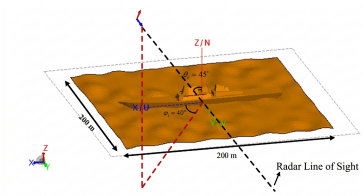

Fig. 5. Definition of elevation (θ_i) and azimuth (ϕ_i) angles in the simulation data.

TABLE II
DATASET DISTRIBUTION ACROSS TARGET CLASSES

Target Class	**Train**	**Val**	**Test**
Warship	472	56	170
Frigate	448	53	161
Containership	474	54	170
1×2 corner reflector	475	57	161
2×2 corner reflector	475	57	161
2×3 corner reflector	475	57	161
Total	2819	337	1014

defined as the angle from the incident wave to the $+z$-axis, and azimuth ϕ_i as the angle of its xy-projection to the $+x$-axis.

TABLE III
ABLATION STUDY OF MODULES

Exp	AGFM	BiRNN	AED	Acc.(%)
Exp-1	✓	×	×	73.87
Exp-2	×	✓	×	72.49
Exp-3	×	×	✓	70.41
Exp-4	✓	✓	×	78.70
Exp-5	✓	×	✓	77.12
Exp-6	×	✓	✓	76.92
Exp-7	✓	✓	✓	**81.56**

TABLE IV
ABLATION STUDY OF ELEVATION (EL) AND AZIMUTH (AZ)

Exp	EL	AZ	Acc.(%)
E1	✓	×	78.80
E2	×	✓	79.29
E3	✓	✓	**81.56**

B. Comparative Experiments

AGFM-Net achieves the best overall accuracy of 81.56%, outperforming all baselines (Table I). It achieves 90.59%, 90.06%, 84.21%, and 88.89% for Container Ship, 1×2, 2×2, and 2×3 reflectors, respectively. Frigate accuracy (65.84%) is marginally below VAE (66.46%) due to strong Warship–Frigate visual similarity. Among baselines, RNN performs weakest, CNN struggles with complex classes, CNN-Bi-RNN shows improvement through spatial-temporal modeling, and VAE excels on Frigate but lacks overall discriminability. AGFM-Net demonstrates clear progression in robustness and inter-class separability.

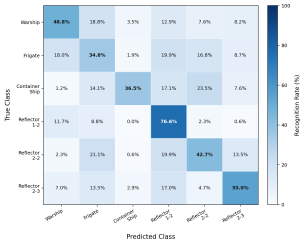

Fig. 6. RNN

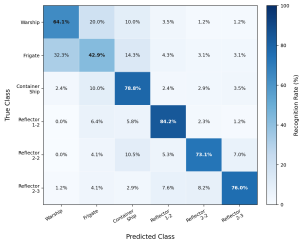

Fig. 7. CNN

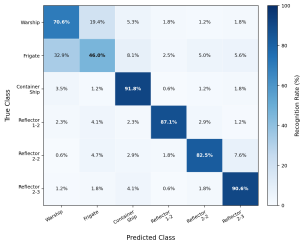

Fig. 8. CNN-Bi-RNN

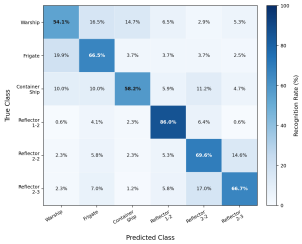

Fig. 9. VAE

C. Ablation Study

Each module contributes independently: AGFM (73.87%), BiRNN (72.49%), and AED (70.41%). Pairwise combinations

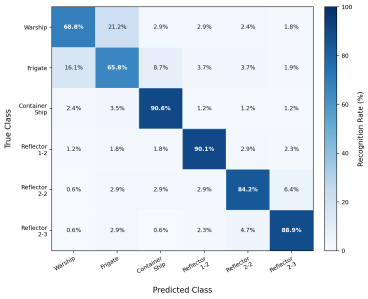

Fig. 10. AGFM-Net

yield further gains (AGFM-Net: 78.70%), and integrating all three achieves the best result (81.56%), confirming synergistic complementarity. Combining both elevation and azimuth angles (E3: 81.56%) outperforms either alone, demonstrating that the two angles provide complementary geometric information effectively utilized by AGFM.

V. Conclusion

This paper proposes AGFM-Net, an angle-guided feature modulation framework for robust HRRP target recognition. The method integrates angle-aware modulation, a dynamic adjustment layer, a BiRNN for sequential modeling, and an attention mechanism for discriminative feature aggregation. Experiments demonstrate consistent superiority over representative baselines with improved robustness and feature separability. The framework is lightweight and easily integrated into existing models. Future work will extend the method to real-world data, multi-target scenarios, and cross-domain tasks.

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